DEMYSTIFYING THE 21ST CENTURY GRANT WORLD

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Presentation Outline

- Personal profile
- Facts about the Granting World
- Writing Fundable Grants
- The DO’s and DON’Ts of the Grant World
- Sustaining the momentum

Profile

- ✓ Associate Prof. of Medical Biochemistry/Director R&I/NACOSTI Board
- ✓PhD in Medical Sciences from HUSM
- ✓Won over 8.5million USD in grants (local & international), and awards
- ✓Recent: USD 4.5 JICA/AMED 2019; TWCF 1.0m, 2019; TWCF 1.2M, 2016; NRF 2017: MKU KES 100M

FACTS ABOUT FUNDERS

- ✓They are (sometimes) desperate to give grants
- ✓They are looking for the right recipient
- ✓They strive to be politically correct

WRITING FUNDABLE GRANTS

- ✓ *Science Communication*
- ✓ Align with the mission, goals of the funding agency
- ✓ Read the seasons & times

❖ Multidisciplinary/multi-institutional=RIGHT PARTNERSHIPS, COLLABORATIONS AND NETWOKS

❖ Community Impact - **Implementation Research (IR)**

IMPLEMENTATION RESEARCH (IR)

- ✓ "IR is the scientific inquiry into questions concerning implementation—can be policies, programmes, or individual practices (collectively called interventions)."

- ✓ IR is especially concerned with the users of the research and not purely the production of knowledge.

Peters DH (2013)
BMJ 2013; 347 doi: https://doi.org/10.1136/bmj.f6753 (Published 20 November 2013)Cite this

WHAT AILS OUR INSTITUTIONS

- ✓No clear policy whether we are *Teaching* or *Research* Universities
- ✓No policies to incentivize researchers (balance between teaching and research)
- ✓No investment in capacity building
- ✓Unnecessary redtape in disbursing of grant funds

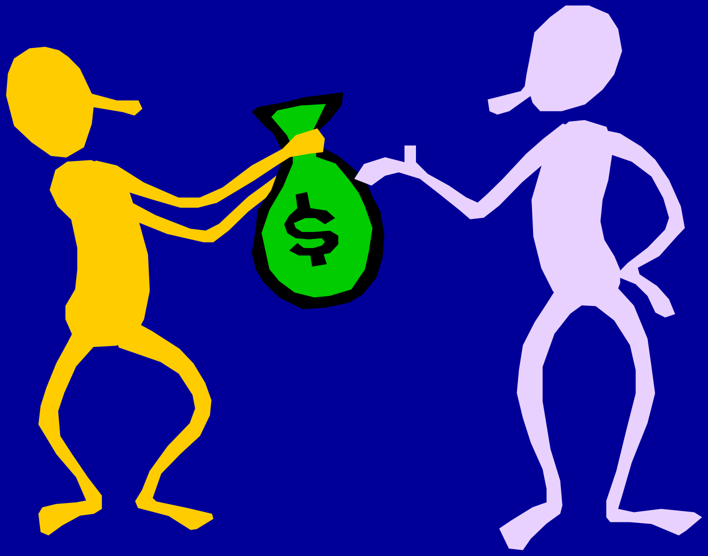
WHAT TO WATCH OUT FOR

- ✓Institutions and Researchers always get shortchanged;
- ✓Partners shortchage by altering collaborators list
- ✓Researchers shortchage institutions ---Overhead charges: Grant officer to review the budget before endorsement

Thank

You

Writing Successful Grant Applications



Dr Jesse Gitaka MBChB, MTM, PhD

References

- Gerin, W., et al., *Writing the NIH Grant Proposal (3rd ed.)*
- Gastel, B. and Day, R. *How to Write and Publish a Scientific Paper (8th ed)*
- Rutkove, R. *Biomedical Research: An Insider's Guide*
- grants.nih.gov/grants/grant_basics.htm
- McGee, R., www.northwestern.edu/climb

Why don't people get funded?

Why don’t people get funded?

- Inadequate concept
- Poor presentation
- Poor understanding of process
- Lack of persistence

An outstanding idea is necessary,
but not enough!

Objectives

- Clarify grant writing process
  - Series of do-able steps
  - Tips to increase likelihood of success
- Discuss common components of grants
  - Sections & content
  - Common mistakes

Caveats

- Many approaches to grant writing
- This is *one* perspective
- Always seek a range of opinions
- **And, always follow the instructions!**

Types of grants

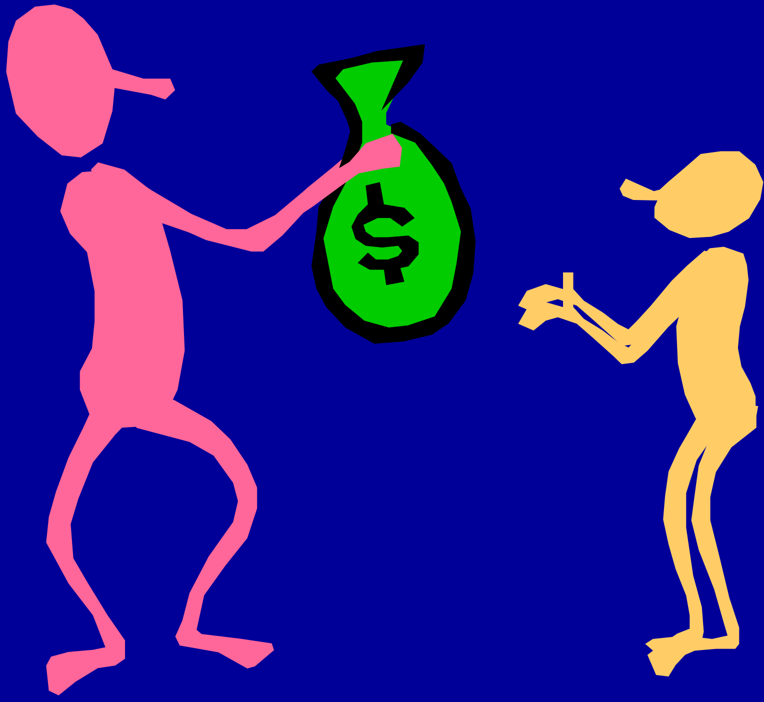
Training & Career Dev.

Research

Conferences

- Clinical trials
- Pilot grants
- Equipment

The process of getting a grant



The process of getting a grant

Preparing

1. Develop concept
2. Identify funding source
   (& reviewers, if can)
4. Talk to program officer
5. Plan backwards
6. Inform others
7. Refine concept

Writing

7. Stock the sections
8. Outline, write, edit
9. Think like a reviewer
10. Get feedback & revise

Submitting

11. Get approvals
12. Submit application
13. Ensure receipt
14. Provide add'l material

Responding (if need be)

15. Await review
16. Study/respond to report

Phase I: preparing

1. Develop concept

2. Identify funding source
3. Talk to program officer
4. Think ahead, plan backwards
5. Inform others
6. Refine concept

1. Develop a concept

That “FITS”

1. Develop a concept that FITS

Fills a gap in knowledge

Important to

- you
- the field
- funding agency

Tests a hypothesis

Short-term investment in long-term goals

SMART

Specific - goals

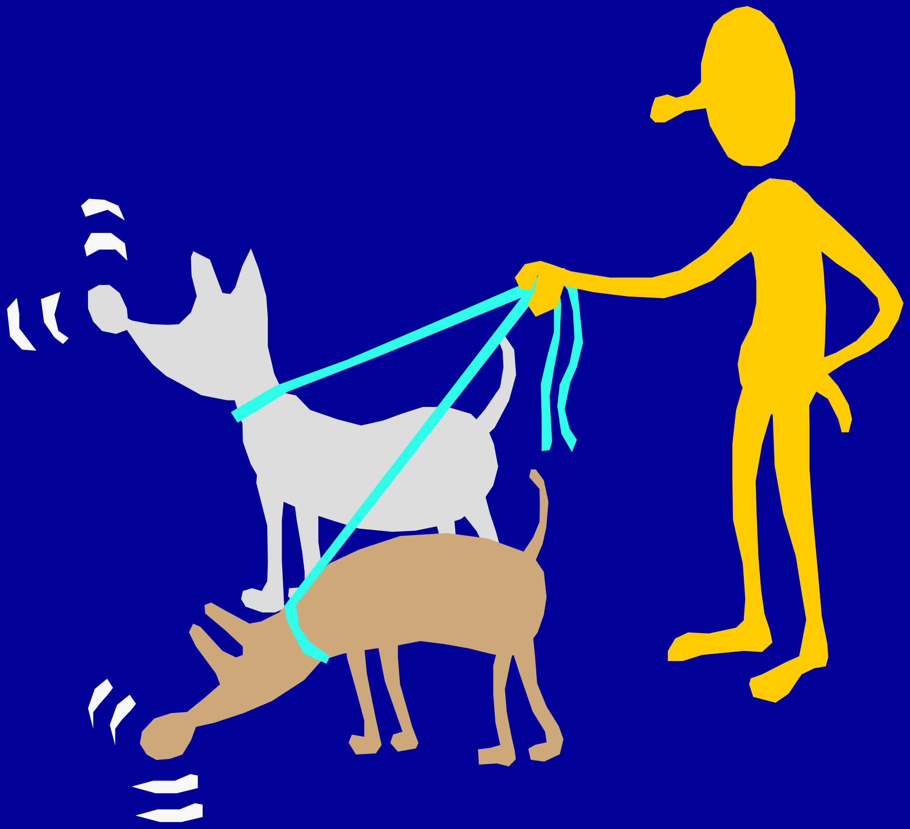
Measurable - outcomes

Achievable - not (too) over-ambitious

Realistic - given resources

Time-limited - in duration

2. Identify funding source



2. Identify funding sources

Where to find information?

- Websites
- Colleagues
- Your institution
- Acknowledgements on papers

*This research was supported in part by grants from the National Institutes of Health (NS039805 and NS060553).

Where to find information?

Fogarty: www.fic.nih.gov

Grantsmanship center:

www.tgci.com/funders

Types of funders

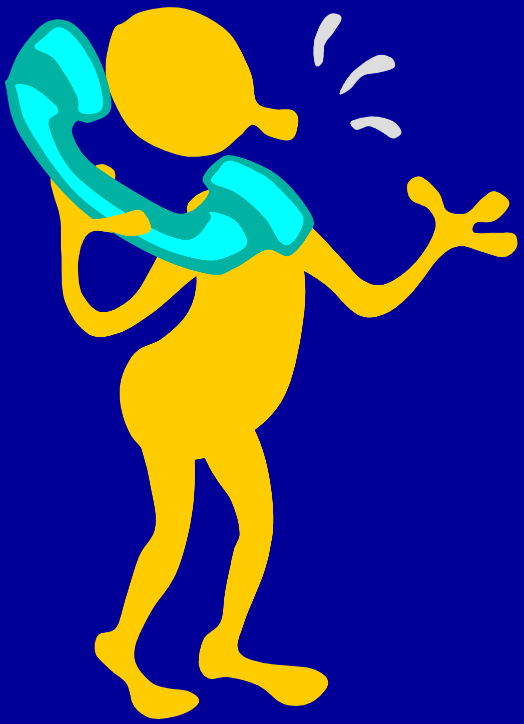
- International
- National
- Private
- Corporations
- Universities
- Professional societies
- Corporations

Match your interests with theirs: Special target groups

- Junior investigators
- Under-represented
- Re-entry
- Special interests
- Investigators from LMICs

3. Talk to Program Officers

…if possible



3. Talk to the program officer

- It can save time and improve proposal!

Ask:

- Is concept relevant to agency mission?
- What sort of funding is available?
  - Amount and duration of funding?
  - Likelihood of funding

4. Think ahead, plan backwards

4. Think ahead, plan backwards

Task	Deadline*
Develop concept	>3 months
Outline, write, edit	2 months
Get approvals	5+ days
Submit application	1-20 days
Grant deadline	0 days

* An example: Adjust for local conditions.

5. Inform those involved

5. Inform those involved

- Individuals at your institution
  - To process application
  - To provide you assistance
- Funding agency: letter of intent
  - Required? (To screen out applicants)
  - Optional? (For internal planning)
- People to give feedback

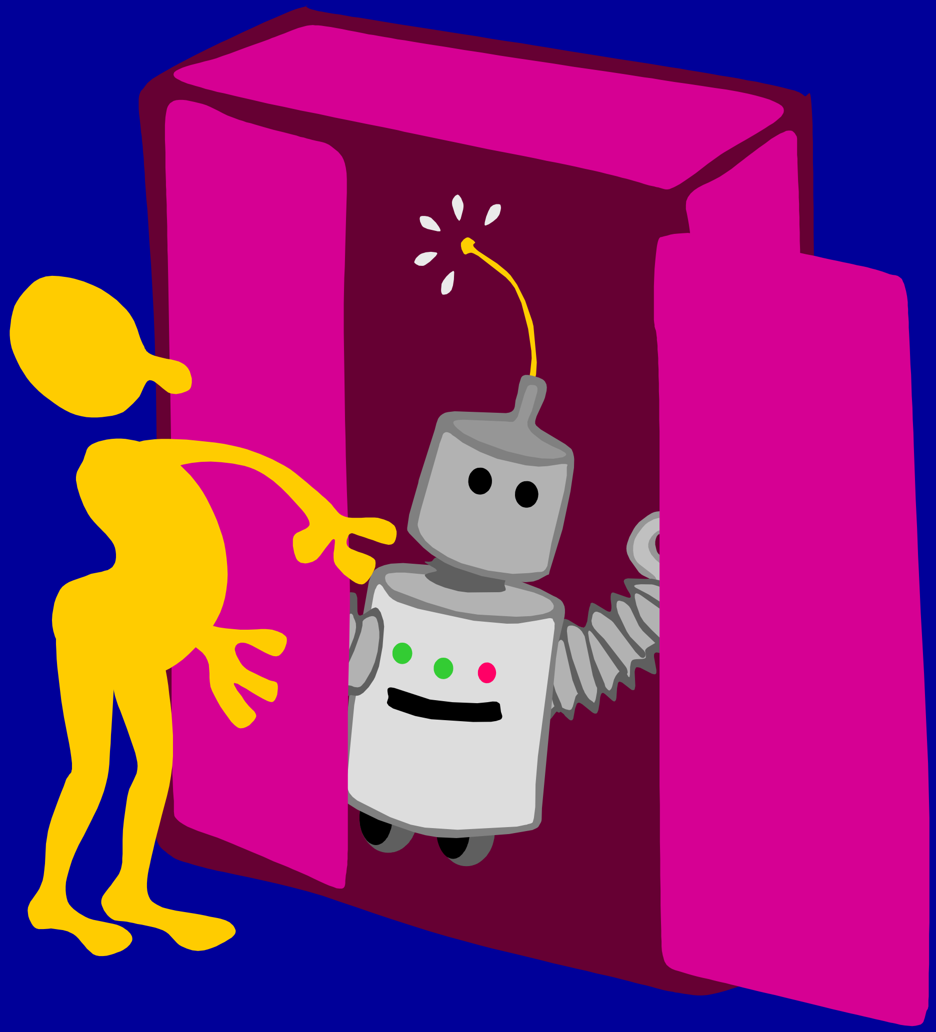
Refine

6. ~~Develop~~ Concept

6. Refine your concept

- Review current literature
- Talk with colleagues
- Think very, very hard

Have a “secret weapon”



Have a “secret weapon”

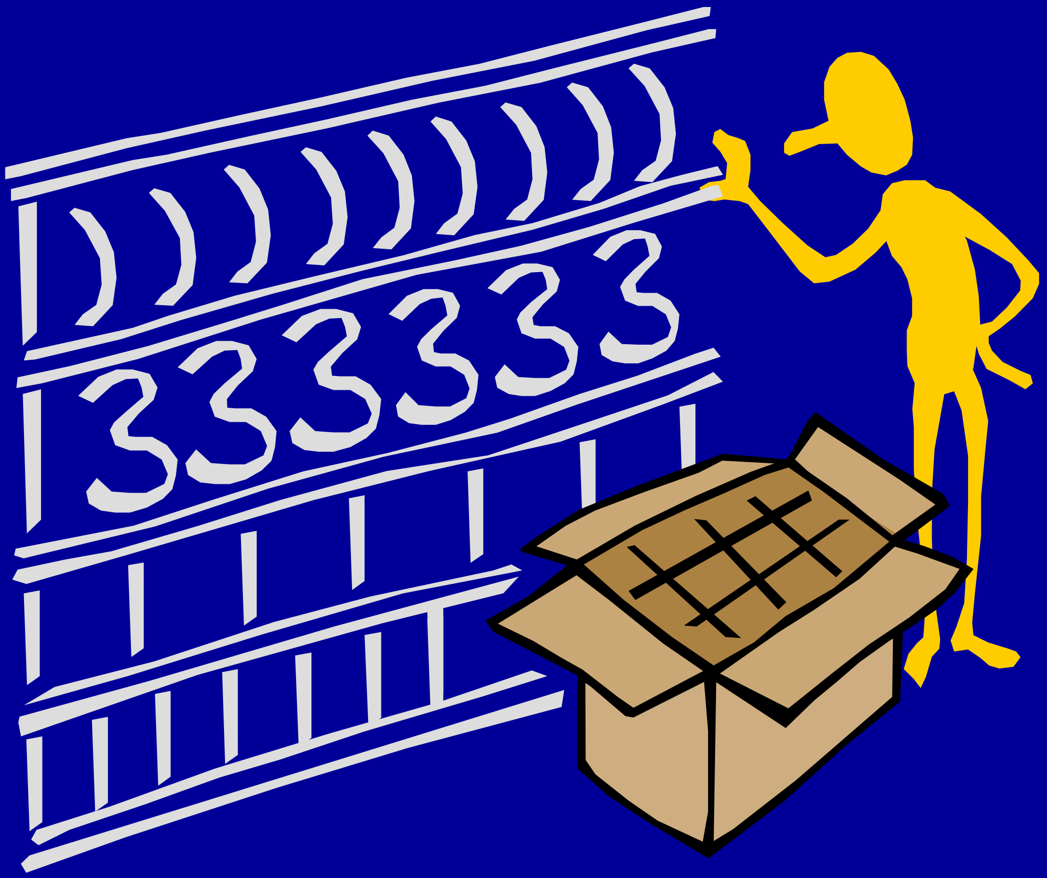
Examples?

- Unusual combination of skills
  - You
  - You + collaborator
- Special method or equipment
- Unique reagent

Phase II: writing the proposal

7. Stock the sections
8. Think like a reviewer
9. Outline, write, edit
10. Get feedback & revise

7. Stock the sections

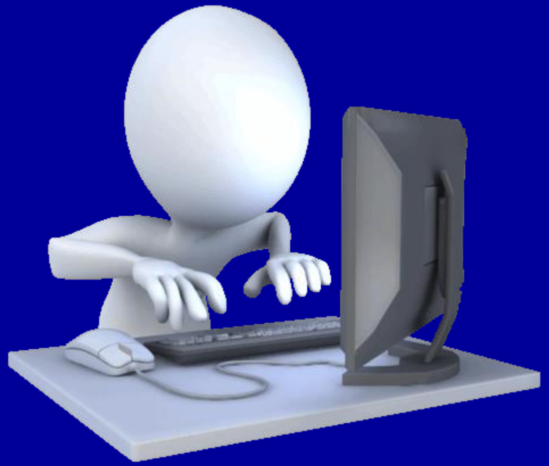


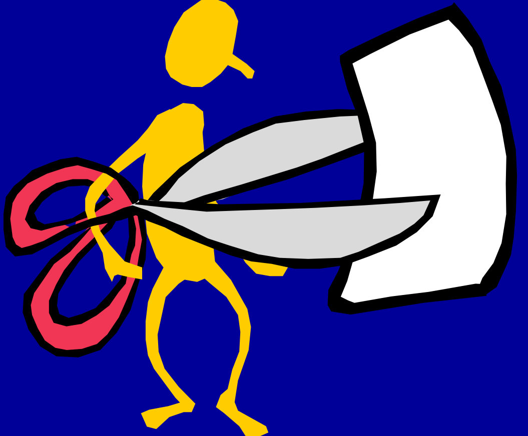
7. Stock the sections

Research Plan:

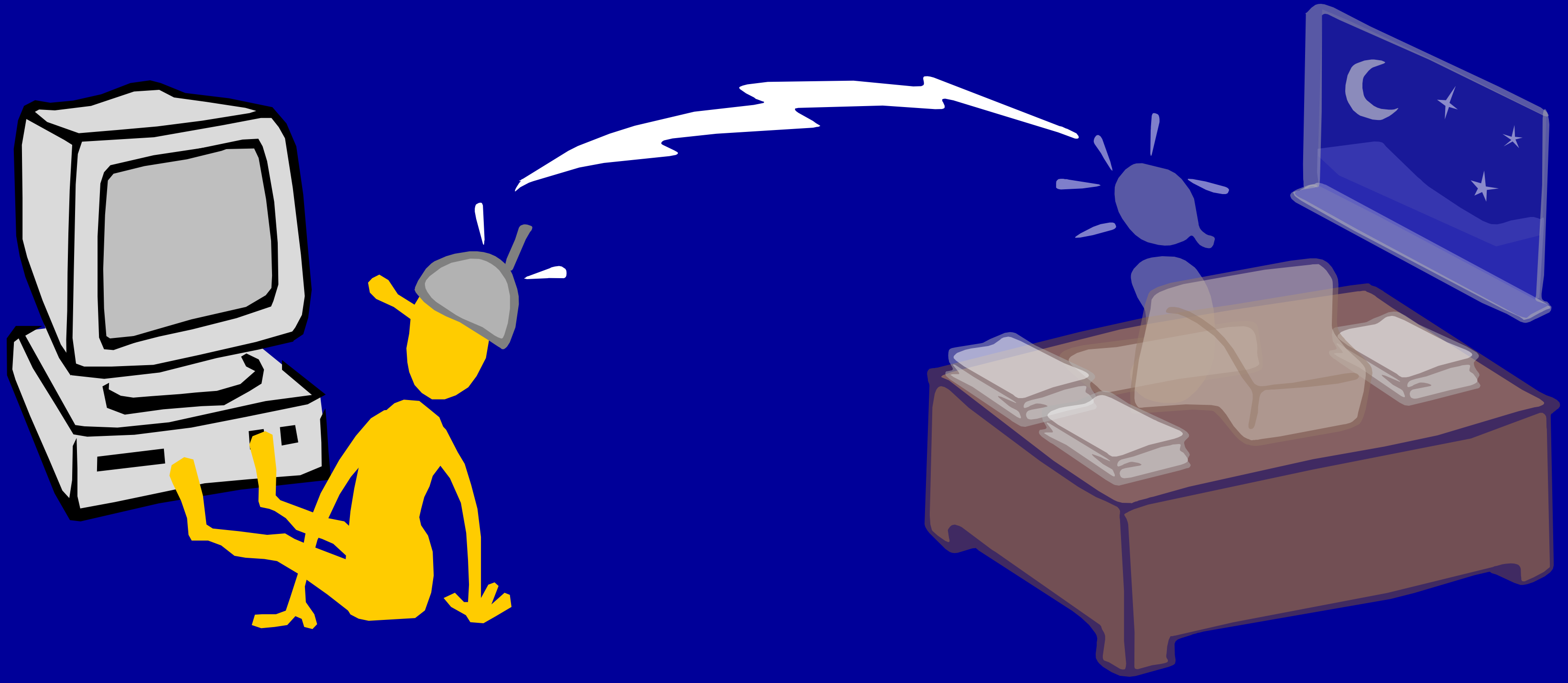
- Objectives (Aims, or Specific Aims)
- Rationale
- Preliminary Data
- Experiments
- References

8. Outline, Write, and Edit





9. Think like a reviewer



NIH/CSR

See online videos

Write for the reviewers

- Anticipate questions, provide answers
- Address review criteria for your proposal
- Standard review criteria (NIH)
  - Significance
  - Innovation
  - Approach
  - Investigator
  - Environment

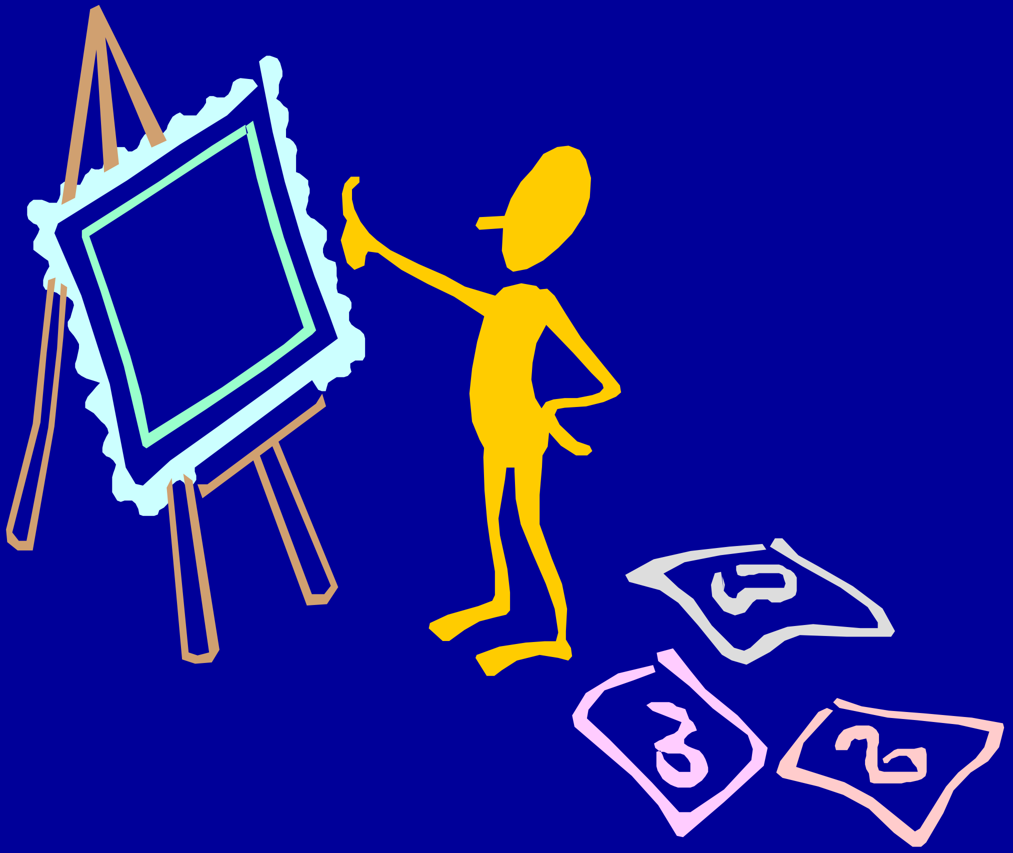
Write for the reviewers

- Use standard organization & format
- State relation to funder’s mission

Good practice

- Write in paragraphs
  - 1 major idea per paragraph
  - Topic sentences
  - Initial paragraphs of section most important
- Make it easy to find key points
  - Use headings frequently
  - Bold face
  - Cross reference
  - Some redundancy

Appearance



Appearance

- Let your text ***b r e a t h e***
  - Indent paragraphs
  - Skip line between paragraphs

3A. Significance

In this section we will first briefly review the literature on stress and its impact on the vulnerability of the brain to injury. Then we will discuss the importance of training in professional skills and in the responsible conduct of research. Finally, we comment on key personnel who will be involved in this effort.

In biological terms, stress typically implies a disequilibrium or deviation from the normal range of an important variable. Stress can trigger physiological and behavioral processes that correct a deviation; it can even strengthen an organism as when preconditioning helps deal with larger challenges ahead. Thus, the biological responses to stress can be lifesaving. However, when the stress response becomes too intense or too long, damage can result. Stress-induced toxicity has been shown in many diseases, including asthma, cancer, multiple sclerosis, and cardiovascular disorders. And it is increasingly clear that stress also plays a role in the pathophysiology of many neuropsychiatric and neurological diseases, including anxiety disorders, depression, schizophrenia, stroke, Alzheimer's disease (AD), and Parkinson's disease (PD).1-6 We propose to examine the processes underlying stress-induced increased vulnerability of CNS to damage, with a particular focus on developmental stress and possible rehabilitation.

Stress can increase the vulnerability of neurons to other insults. Experimental evidence indicates that stress can render neurons more vulnerable to subsequent insult or exacerbate damage incurred from insult. Robert Sapolsky has referred to this as "neuroendangerment".7 For example, prolonged physical and/or psychosocial stress leads to psychopathological symptoms and cerebral or hippocampal atrophy in humans.8,9 In addition, hypercortisolemiais associated with increased neurological and neuropathogical deficits and unfavorable prognoses after stroke.10,11 Stress also has been shown to correlate with the development of new brain lesions in patients with multiple sclerosis.12 Additionally, a combination of moderate physical and social stresses has been shown to exacerbate clinical symptoms in experimental autoimmune encephalomyelitis, an animal model of multiple sclerosis.13 Finally, it has been proposed that stress can exacerbate or even precipitate precipitate Parkinson's.14-16

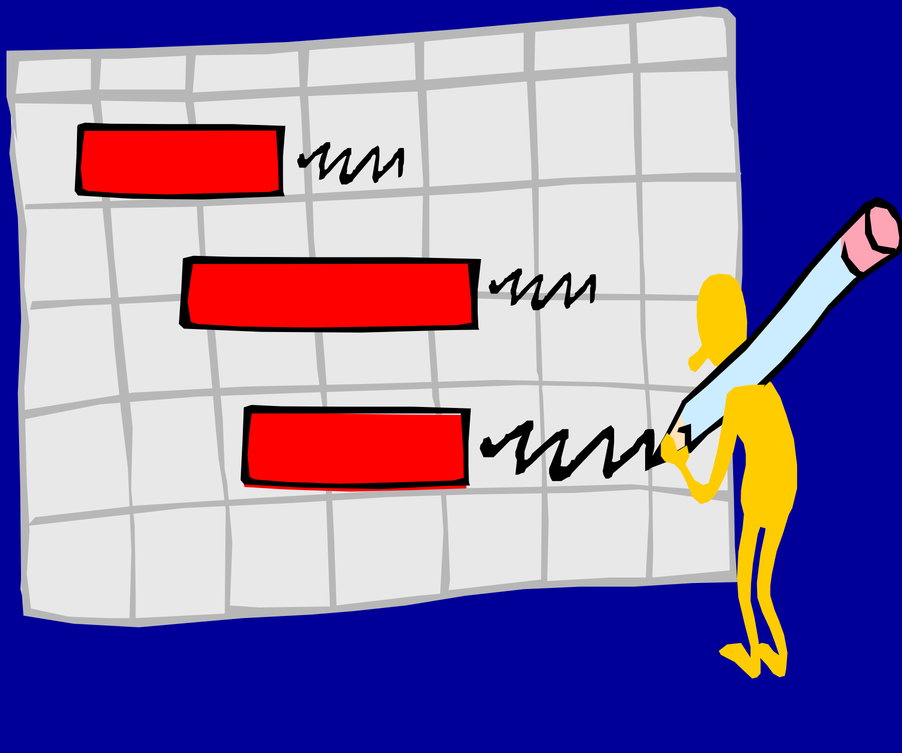
3A. SIGNIFICANCE

In this section we will first briefly review the literature on stress and its impact on the vulnerability of the brain to injury. Then we will discuss the importance of training in professional skills and in the responsible conduct of research. Finally we will comment on the key personnel who will be involved in this effort.

In biological terms, *stress* typically implies a disequilibrium or deviation from the normal range of an important variable. Stress can trigger physiological and behavioral processes that correct a deviation; it can even strengthen an organism as when preconditioning helps deal with larger challenges ahead. Thus, the biological responses to stress can be lifesaving. However, when the stress response becomes too intense or too long, damage can result. Stress-induced toxicity has been shown in many diseases, including asthma, cancer, multiple sclerosis, and cardiovascular disorders. And it is increasingly clear that stress also plays a role in the pathophysiology of many neuropsychiatric and neurological diseases, including anxiety disorders, depression, schizophrenia, stroke, Alzheimer's disease (AD), and Parkinson's disease (PD) (see reviews by Natelson, 1983; Anisman et al., 1985; Anisman & Zacharko, 1990; Deutch, 1993; Fibiger, 1995; Raber, 1998). ***We propose to examine the processes underlying stress-induced increased vulnerability of CNS to damage, with a particular focus on developmental stress and possible rehabilitation.***

Stress can increase the vulnerability of neurons to other insults. Experimental evidence indicates that stress can render neurons more vulnerable to

10. Get Feedback and Revise



10. Get feedback and revise

From whom?

- Mentors
- Colleagues
- Former reviewers
- People with good English skills

10. Get feedback and revise

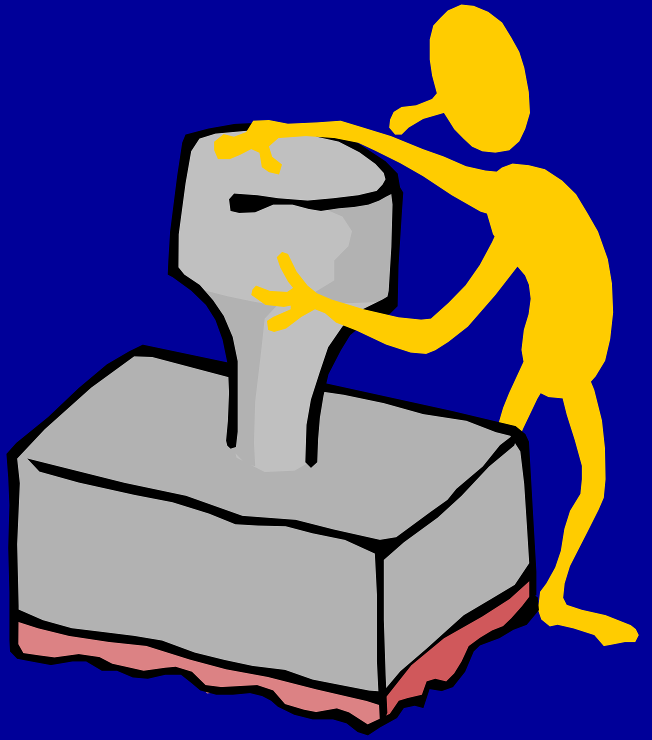
About what?

- Overall application
- Aims
- Research plan
- Language
- Appearance

Phase III: submitting

11. Get approvals
12. Submit application
13. Ensure receipt
14. Provide additional material

11. Get approvals



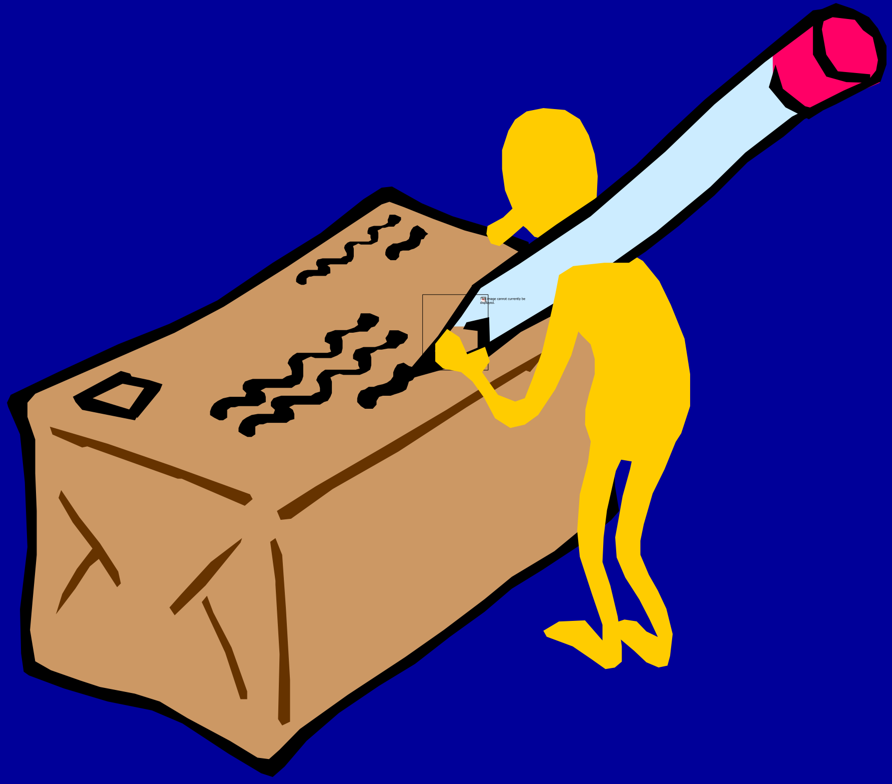
12. Submit application

12. Submit application

- Know the deadline
- Keep time zone in mind
- Anticipate problems
- Give yourself extra time

13. Ensure receipt

14. Provide additional material
(if permitted)

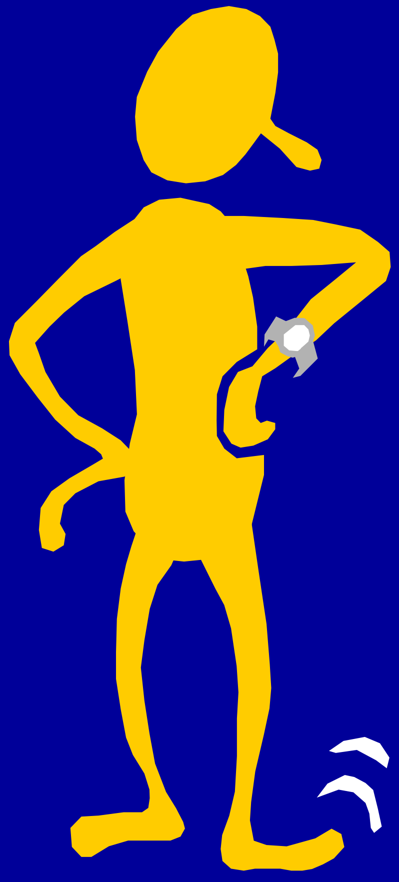


- Updated credentials, accomplishments
- Accepted papers

Phase IV: responding

15. Await review
16. Study/respond to report

15. Await review



15. Await review

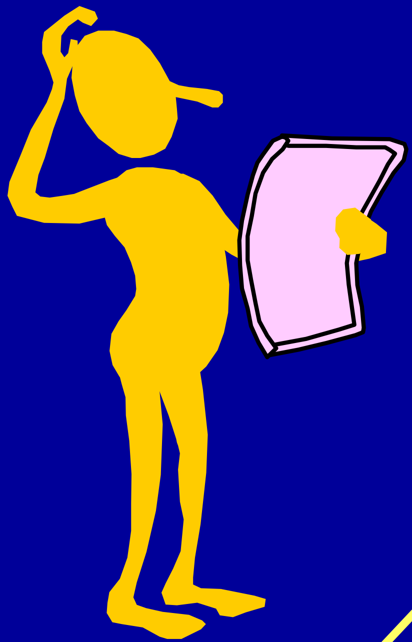
What will be happening:

1. Assignment
2. Evaluation
3. Preparation of report

Warning: the report may

- be incomplete
- contain contradictions
- contain errors

16. Study report and respond



Reasons for rejection

- Unoriginal ideas
- Diffuse, superficial
- Lack of knowledge
- Uncertain future directions
- Inadequate rationale
- Poor reasoning
- Unrealistic workload
- Lack of expt'l detail
- Uncritical approach

Killer words

"Overambitious"

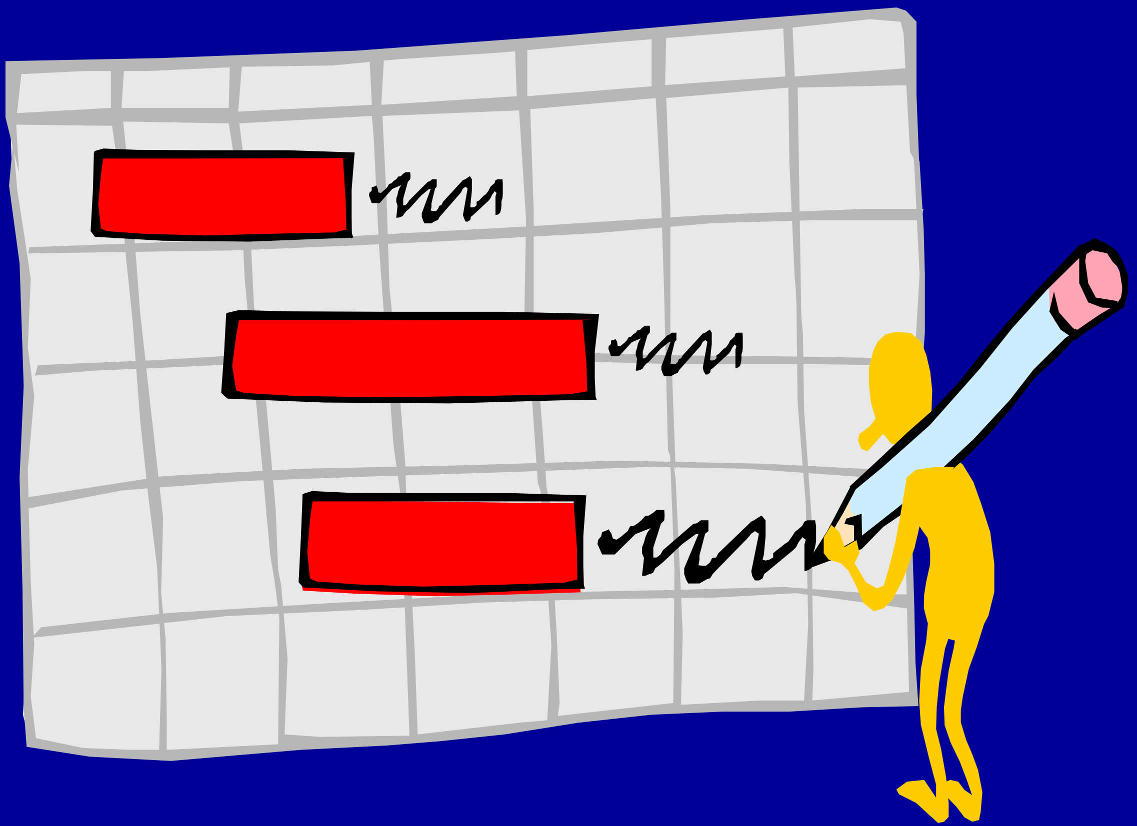
"Descriptive"

"Superficial"

Standard components of a grant application

1. Title
2. Abstract
3. Budget & justification
4. Research plan
5. Subject welfare
6. Biographical information on key personnel
7. Supplementary materials

Research Plan



Research plan

Sections

1. Objectives (Specific Aims)*

2. Rationale, background, (Significance; Innovation)

3. Preliminary studies (for NIH usually part of Significance or Res. Design)

4. ~~Research Design~~ (Approach)

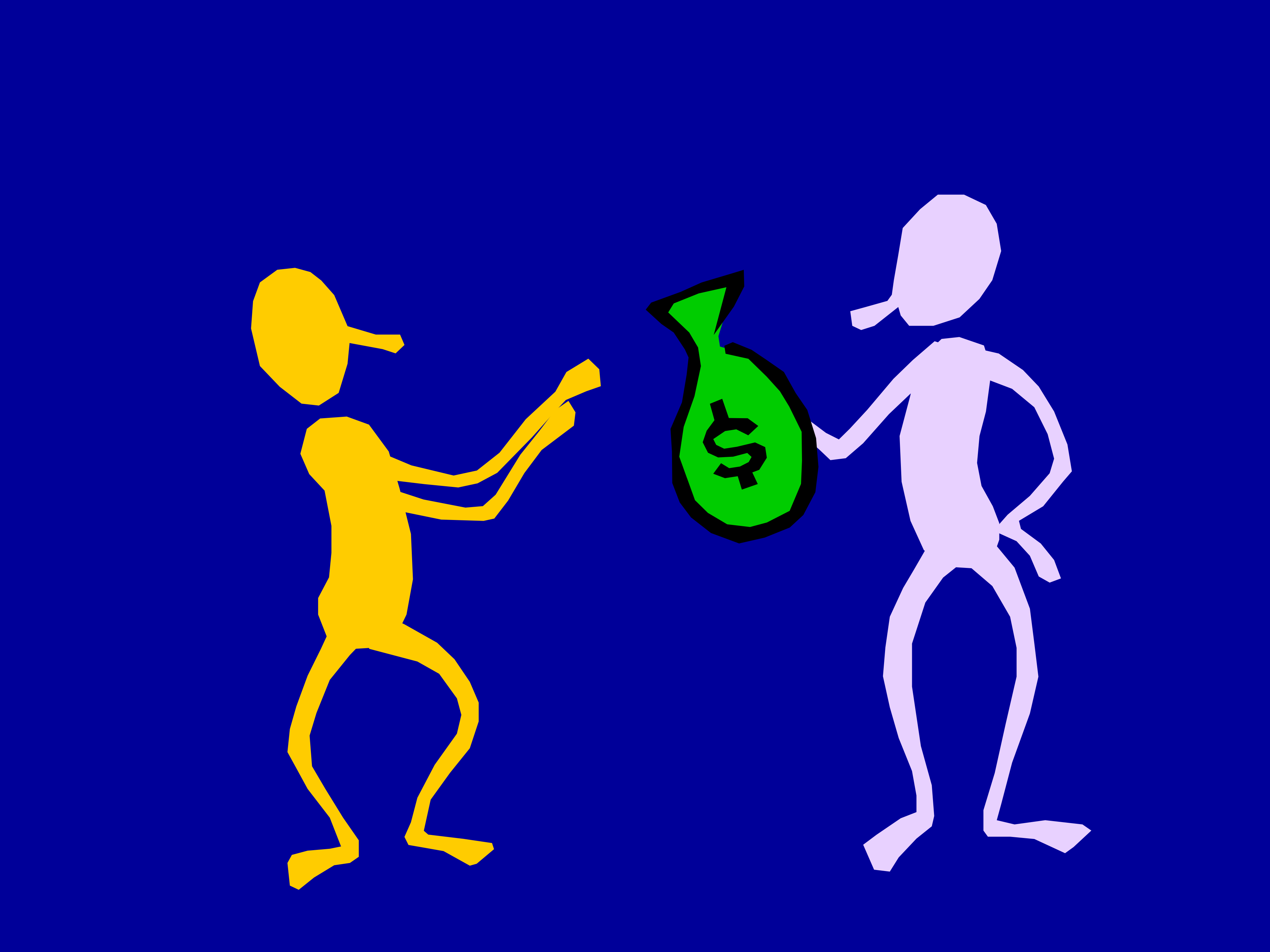
Focus

- What?
- Why? New??
- Feasible?
- How?

* NIH terminology is indicated in parentheses.

Summary

- There is money available
- May need to start small
- Getting the funds takes
  - A good idea
  - A proper match
  - Good grantspersonship
  - Persistence
- It is hard (& sometimes frustrating) work
- It is absolutely worth it!



Thanks to Beth and Michael
and NIH

Thank you

Questions

PRACTICAL OUTLOOK

- TWCF Grants (Muregi)
- County Innovation Grant: the Bungoma experience (Gitaka)
- Science Communication Grant (Muregi)
- Interdisciplinary research for an integrated community-directed strategy to eliminate malaria in tropical Africa (Gitaka)

PRACTICAL OUTLOOK

Participatory Session

(Q & A)

Thank You